

*** Veri Hazırlığı Tamamlandı! Toplam Analiz Edilen Satır: 753

--- Eksik Veri Kontrolü ---

CharacteristicName

pH 0

Sulfate 0

Potability 0

dtype: int64

--- İstatistiksel Özet ---

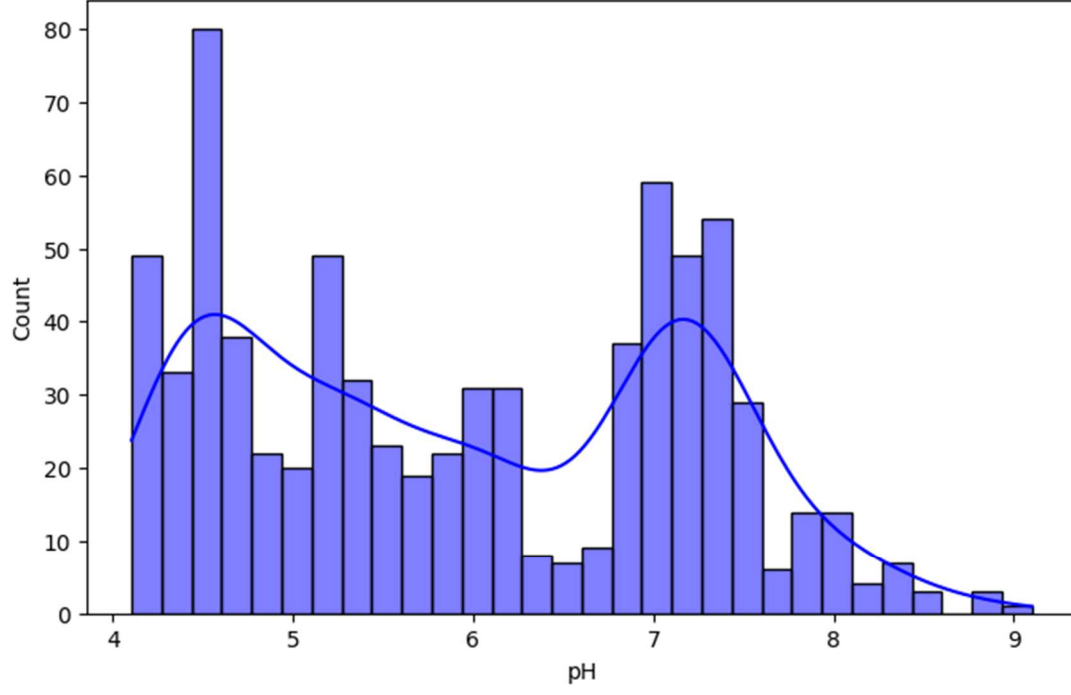
CharacteristicName	pH	Sulfate	Potability
count	753.000000	753.000000	753.000000
mean	5.947244	14.556985	0.379814
std	1.239114	88.201719	0.485663
min	4.105000	1.100000	0.000000
25%	4.700000	2.200000	0.000000
50%	5.865000	4.100000	0.000000
75%	7.100000	7.000000	1.000000
max	9.100000	1265.000000	1.000000



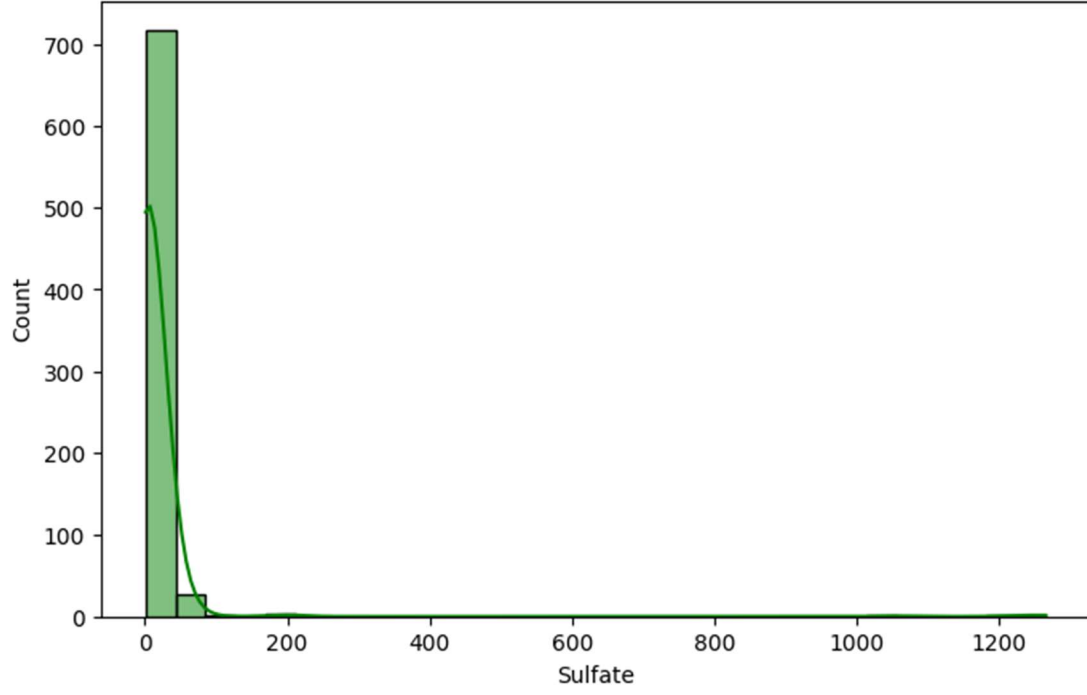
--- Sınıflara Göre Ortalama Değerler ---

CharacteristicName	pH	Sulfate
Potability		
0	5.135821	13.518701
1	7.272191	16.252366

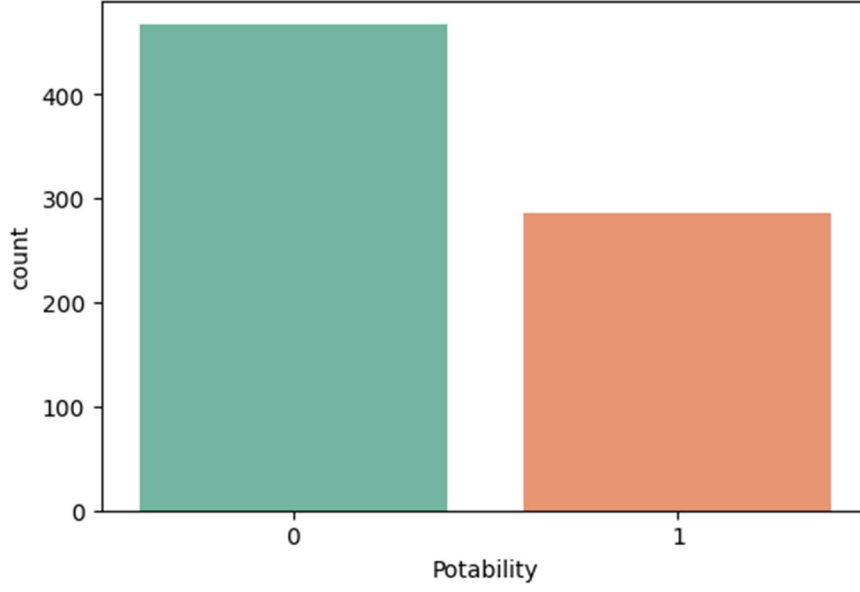
1. pH Değerlerinin Dağılımı



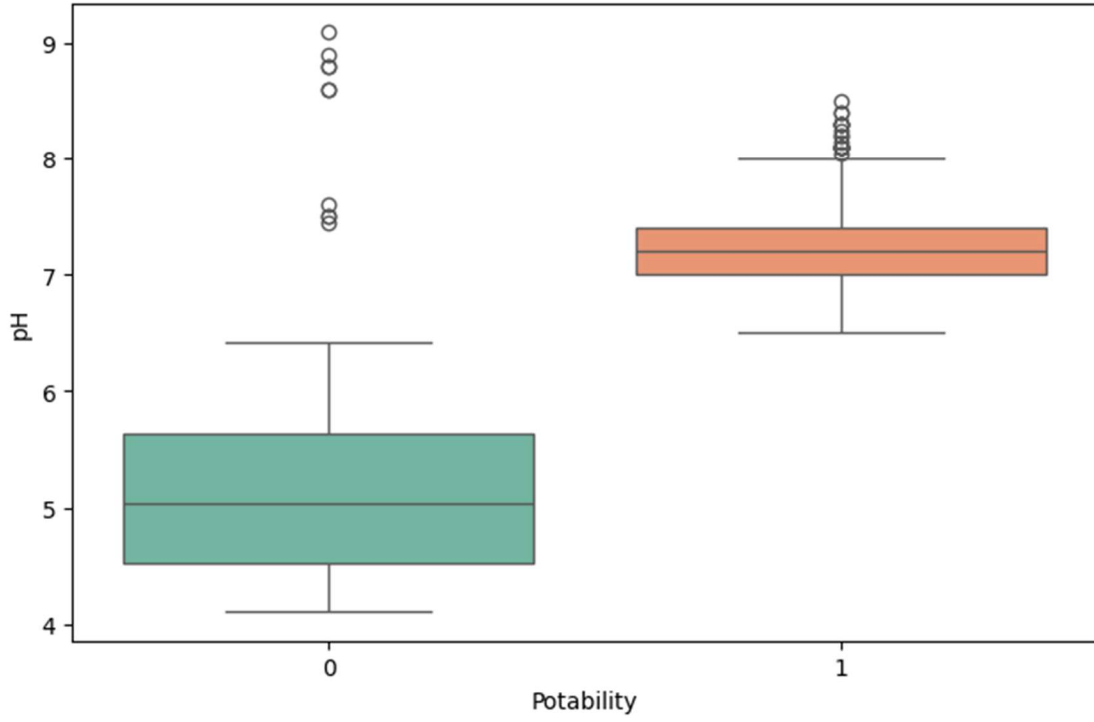
2. Sülfat Değerlerinin Dağılımı



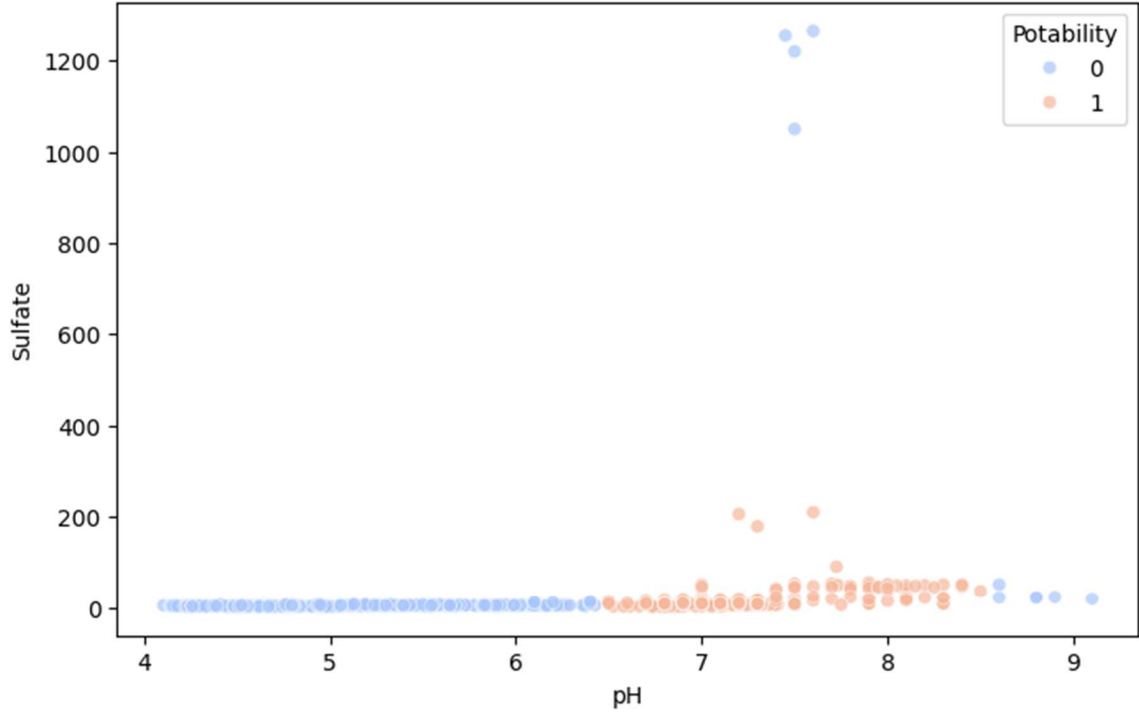
3. İçilebilirlik Durum Dağılımı (0: Hayır, 1: Evet)



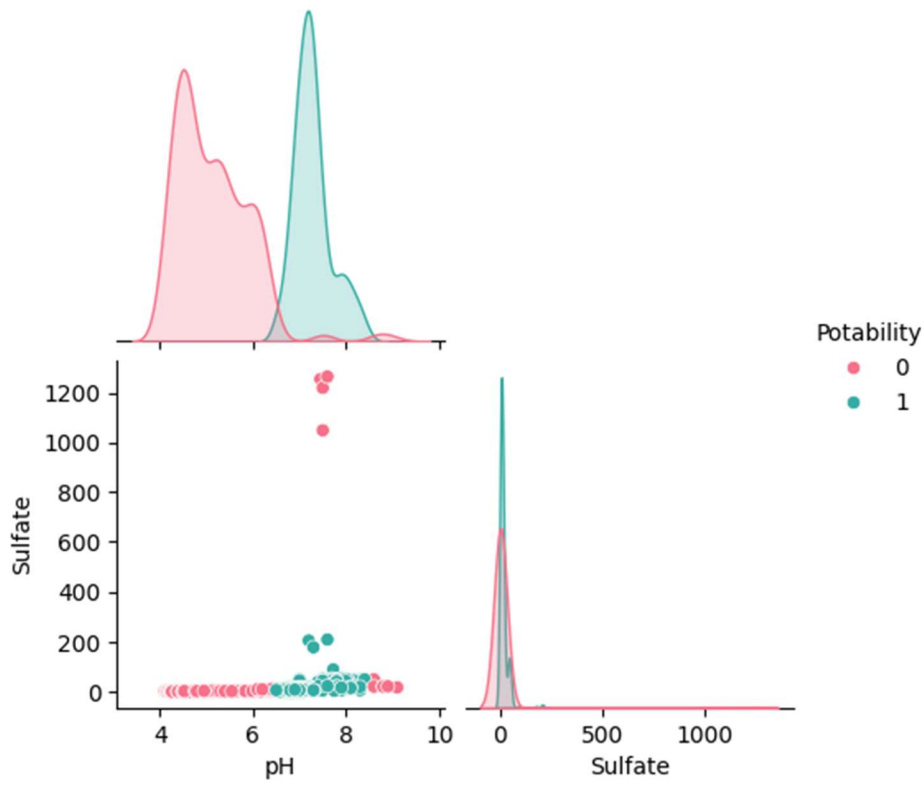
4. İçilebilirlik Durumuna Göre pH Kutu Grafiği (Boxplot)



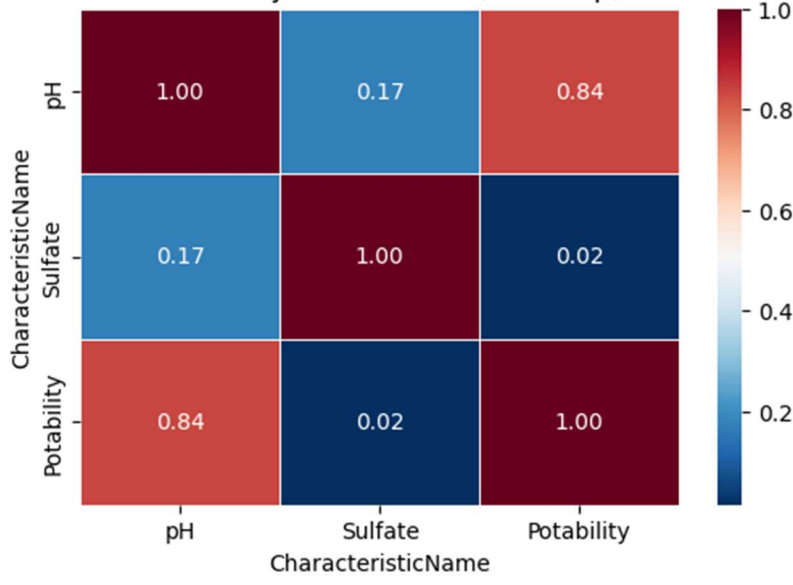
5. Dağılım Grafiği: pH vs Sülfat



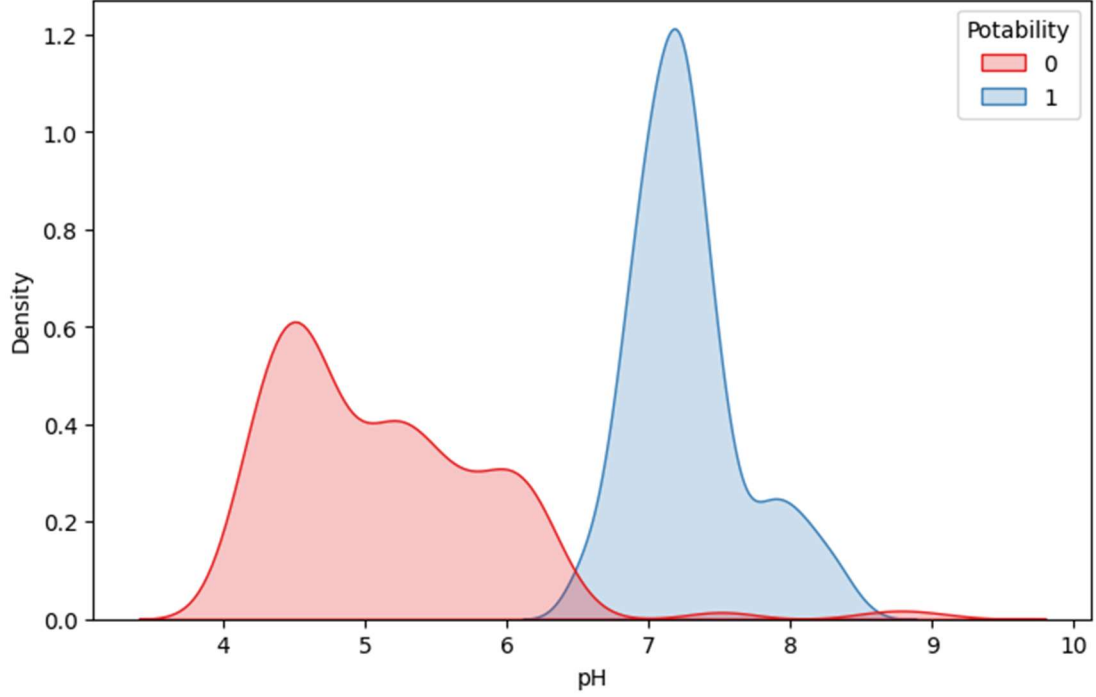
6. İkili ilişkiler (Pairplot)



7. Korelasyon Isı Haritası (Heatmap)



8. İçilebilirlik Durumuna Göre pH KDE (Yoğunluk) Grafiği



Makine Öğrenmesi Modelleri Eğitiliyor...

--- Logistic Regression Sonuçları ---

Accuracy : 0.987

Precision: 0.966

Recall : 1.000

F1-score : 0.983

ROC-AUC : 0.979

--- Random Forest (Baseline) Sonuçları ---

Accuracy : 1.000

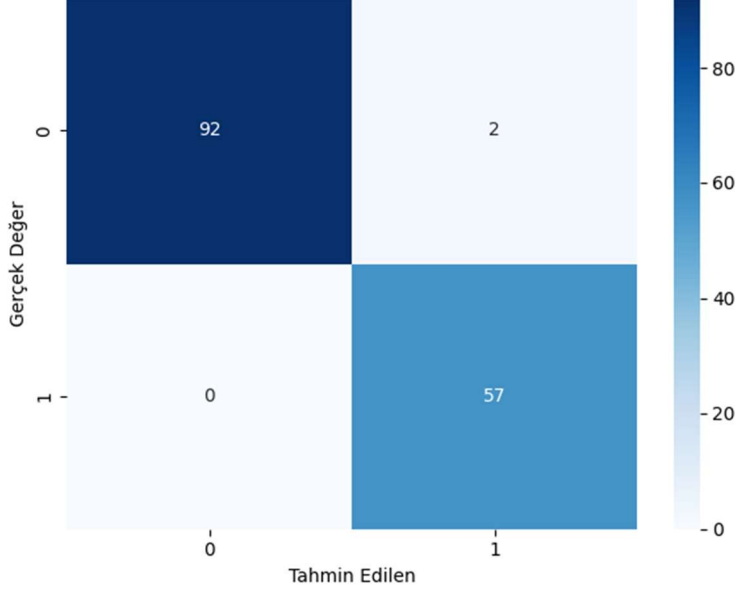
Precision: 1.000

Recall : 1.000

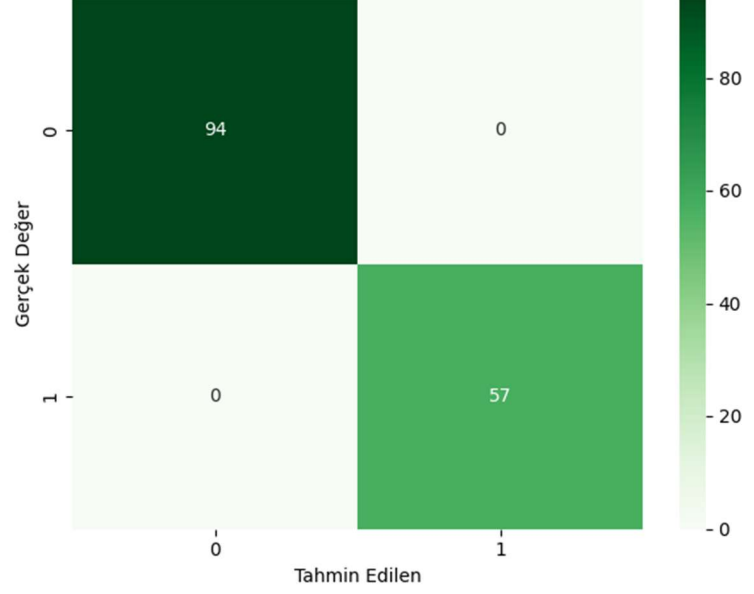
F1-score : 1.000

ROC-AUC : 1.000

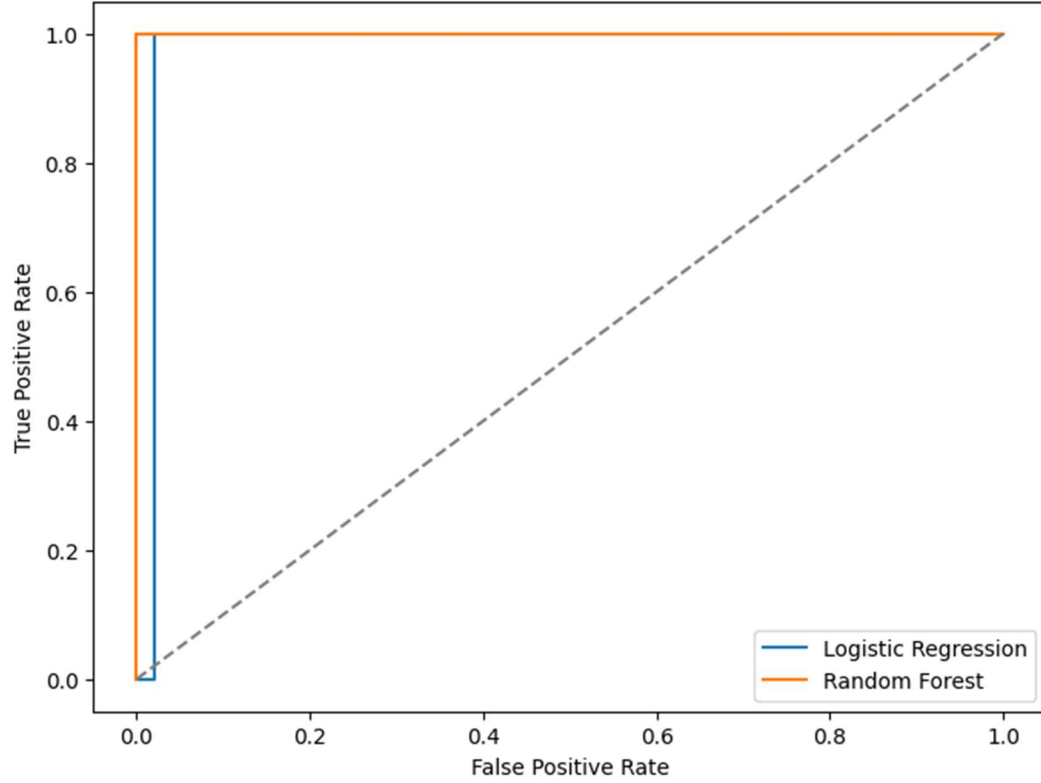
Logistic Regression Confusion Matrix



Random Forest Confusion Matrix



ROC Curve Karşılaştırması



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NİHAİ İYİLEŞTİRİLMİŞ MODEL (PART B) ÇALIŞTIRILIYOR...
=====
--- Optimized Pipeline (SMOTE + Random Forest) Sonuçları ---
Accuracy : 1.000
Precision: 1.000
Recall   : 1.000
F1-score : 1.000
ROC-AUC  : 1.000
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Final Model Confusion Matrix (Orijinal EPA Verisi)

